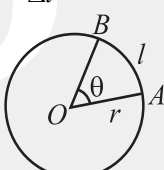
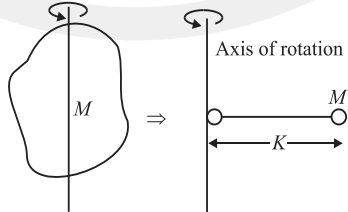
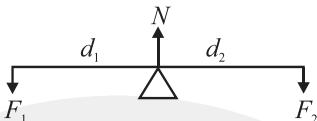
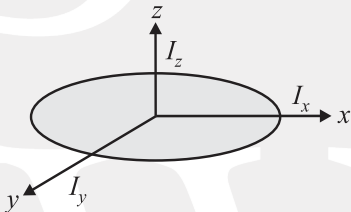
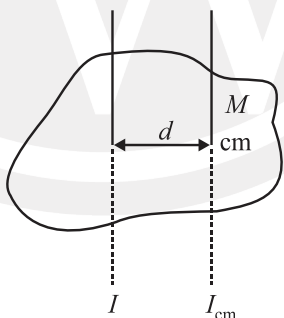


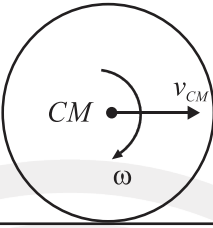
Rotational Motion

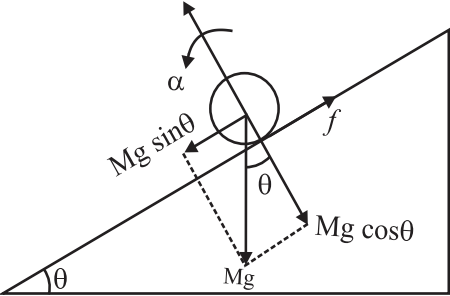
Sr. No.	Concept	Formulas	Other information
1.	Terms related to rotational motion	Angular displacement (θ) = $\frac{\text{arc length } (l)}{\text{radius } (r)}$ Angular velocity = $\frac{\Delta\theta}{\Delta t}$  Instantaneous angular velocity is: $\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$ Relation between angular velocity and linear velocity: $\vec{v} = \vec{\omega} \times \vec{r}$ Angular acceleration (α) = $\frac{d\omega}{dt} = \frac{d^2\theta}{dt^2} = \omega \frac{d\omega}{d\theta}$	$d\theta$ = small angular displacement. dt = small time interval. r = radius of the circle.
2.	Particle moving in circle with uniform speed.	Angular velocity: $\omega = \frac{2\pi N}{t}$ where $\theta = 2\pi N$ If T is time period of revolution and n is frequency, then $\omega = \frac{2\pi}{T} = 2\pi n$	N = number of revolutions in t second. θ = angular displacement.
3.	Equations of rotational motion	Equations of rotational motion can be written as: (i) $\omega = \omega_0 + \alpha t$ (ii) $\theta = \omega_0 t + \frac{1}{2} \alpha t^2$ (iii) $\omega^2 - \omega_0^2 = 2\alpha\theta$ (iv) $\theta_{n^{th}} = \omega_0 + \frac{\alpha}{2} (2n - 1)$	ω_0 = initial angular velocity ω = final angular velocity α = angular acceleration. θ = angular displacement after the time t. $\theta_{n^{th}}$ = angular displacement in n^{th} second.

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4.	Torque (τ)	Torque, $\tau = rF \sin \theta$ or $\vec{\tau} = \vec{r} \times \vec{F}$	F = Force acting on the body. $\vec{r}$ = position vector of particle. θ = angle between $\vec{F}$ and $\vec{r}$
5.	Angular momentum	Angular momentum, $L = rp \sin \theta$ or $\vec{L} = \vec{r} \times \vec{p}$ A particle moving with uniform speed in a circle having angular momentum: $L = mvr$	p = linear momentum $\vec{r}$ = position vector of particle. θ = angle between $\vec{p}$ and $\vec{r}$.
6.	Relation between torque, moment of inertia and angular momentum	(i) Torque = Rate of change of angular momentum or $\tau = \frac{dL}{dt}$ (ii) $\tau = I\alpha$ (iii) $L = I\omega$	τ = torque. I = moment of inertia. α = angular acceleration.
6.	Moment of inertia (I)	Moment of inertia of a body about the given axis of rotation, $I = m_1 r_1^2 + m_2 r_2^2 + \dots + m_n r_n^2 = \sum_{i=1}^n m_i r_i^2.$ For a single particle: $I = mr^2$ For a continuous object: $I = \int dI = r^2 \int dm$	dm = small mass element m = mass of the single particle r = perpendicular distance from the axis.
7.	Radius of gyration (K)	If a body has mass M and radius of gyration K , then moment of inertia, $I = MK^2 \Rightarrow K = \sqrt{\frac{I}{M}}$  When all the particles having same mass (for system of particles): $K = \sqrt{\frac{r_1^2 + r_2^2 + r_3^2 + \dots + r_n^2}{n}}$	$r_1, r_2, r_3, \dots, r_n$ = perpendicular distances of particles from axis of rotation. n = number of particles. M = mass of the body

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8.	Equilibrium of a rigid body	Translational equilibrium: $\vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_n = \sum_{i=1}^n \vec{F}_i = 0$ Rotational equilibrium: $\vec{\tau}_1 + \vec{\tau}_2 + \dots + \vec{\tau}_n = \sum_{i=1}^n \vec{\tau}_i = 0$	$F_1, F_2, \dots, F_n$ = the working forces $\tau_1, \tau_2, \dots, \tau_n$ = the torques acting on the body.
9.	Principle of moments	For translational equilibrium. $N - F_1 - F_2 = 0$  For rotational equilibrium: or, $d_1 F_1 - d_2 F_2 = 0$	F_1 and F_2 = two forces acts on two livers of the body. d_1 and d_2 = distance from the fulcrum
10.	Perpendicular and parallel axis theorem	Perpendicular Axis Theorem: Only applicable to plane lamina (that means for 2-D objects only) $I_z = I_x + I_y$ (when object is in $x - y$ plane).  Parallel Axis Theorem: (Applicable to any type of object): $I = I_{cm} + Md^2$ 	I_{cm} = moment of inertia of the body about centre of mass. M = total mass of the body. d = distance of the parallel axis from centre of mass.
11.	Moment of inertia of a ring (M)	M.I. of a circular ring about an axis through its centre and perpendicular to its plane, $I = MR^2$ M.I. of a thin ring about any diameter, $I = \frac{1}{2} MR^2$ M.I. of a thin ring about any tangent in its plane, $I = \frac{3}{2} MR^2$	M = mass of the ring. R = radius of the ring.

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12.	Moment of inertia of a disc (M)	M.I. of a circular disc about an axis through its centre and perpendicular to its plane, $I = \frac{1}{2}MR^2$ M.I. of a circular disc about any diameter, $I = \frac{1}{4}MR^2$ M.I. of a circular disc about a tangent in its plane, $I = \frac{5}{4}MR^2$	M = mass of the disc. R = radius of the disc.
13.	Moment of inertia of a rod (M)	M.I. of a thin rod about an axis through its middle point and perpendicular to rod, $I = \frac{1}{12}ML^2$ M.I. of a thin rod about an axis through its one end and perpendicular to rod, $I = \frac{1}{3}ML^2$	M = mass of the thin rod. L = length of the rod.
14.	Moment of inertia of a rectangular lamina (M)	M.I. of a rectangular lamina about an axis through its centre and perpendicular to its plane, $I = M \left(\frac{l^2 + b^2}{12} \right)$	l = length of the lamina. b = breadth of the lamina.
15.	Moment of inertia of a cylinder (M)	M.I. of a circular solid cylinder about its symmetry axis, $I = \frac{1}{2}MR^2$ M.I. of a circular hollow cylinder about its axis $I = MR^2$	M = Mass of the cylinder in each case. R = radius of the cylinder in each case.
16.	Moment of inertia of a solid sphere (M)	M.I. of a solid sphere about an axis through its centre, $I = \frac{2}{5}MR^2$ M.I. of a solid sphere about any tangent, $I = \frac{7}{5}MR^2$	M = mass of the solid sphere. R = radius of the solid sphere.

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17.	Moment of inertia of a hollow sphere (M)	M.I. of a hollow sphere about an axis through its centre, $I = \frac{2}{3}MR^2$ M.I. of a hollow sphere about any tangent, $I = \frac{5}{3}MR^2$	M = mass of the hollow sphere. R = radius of the hollow sphere.
18.	Kinetic energy of a rotating body (pure rolling)	 Rotational kinetic energy $= \frac{1}{2}I\omega^2$ $\therefore$ Kinetic energy of rolling motion, $E = \frac{1}{2}I\omega^2 + \frac{1}{2}Mv_{CM}^2$ Substituting the value of $I = MK^2$, where K is radius of gyration and $v_{CM} = R\omega$ we get $E = \frac{1}{2}MK^2\left(\frac{v_{CM}}{R}\right)^2 + \frac{1}{2}Mv_{CM}^2$ $\Rightarrow E = \frac{1}{2}Mv_{CM}^2\left(1 + \frac{K^2}{R^2}\right)$	I = moment of inertia. ω = angular velocity. M = mass of the body v_{CM} = linear velocity of centre of mass of the body. K = rotational kinetic energy
19.	Work energy theorem in rotational motion	Work done by the torque = change in kinetic energy of rotation $W = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$	W = work done ω_2 = final angular velocity ω_1 = initial angular velocity
20.	Instantaneous power (P) associated with torque	$P = \frac{dW}{dt} = \frac{d}{dt}(\tau\theta) = \tau \frac{d\theta}{dt} = \tau\omega$ As the power is a scalar quantity, it is written as $P = \vec{\tau} \cdot \vec{\omega}$	τ = torque acting on the body. ω = angular velocity.
21.	Law of conservation of angular momentum	In the absence of any external torque, $L = I\omega = \text{constant}$ or $I_1\omega_1 = I_2\omega_2$ $I_1 \frac{2\pi}{T_1} = I_2 \frac{2\pi}{T_2}$	L = angular momentum. I = moment of inertia. ω = angular velocity.

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22.	Motion of a cylinder rolling down an inclined plane without slipping	 (i) Linear acceleration of the cylinder down the inclined plane, $a = \frac{g \sin \theta}{1 + \frac{I}{MR^2}} = \frac{2}{3} g \sin \theta \quad \left[\text{for cylinder } I = \frac{MR^2}{2} \right]$ (ii) Force of friction between inclined plane and cylinder, $f = \frac{1}{3} Mg \sin \theta$ (iii) Condition for rolling of cylinder without slipping is $\frac{1}{3} \tan \theta \leq \mu_s$ where μ_s is the coefficient of static friction.	θ = angle of inclination I = moment of inertia of the body. M = mass of the body. f = force of friction. α = angular acceleration.